PWANI UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE

HEALTHLINK:
AN INTELLIGENT AI-POWERED TELEMEDICINE PLATFORM

CONCEPT PAPER

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Document Information

Field	Details
Project Title	HealthLink: An Intelligent AI-Powered Telemedicine Platform with Integrated Mobile Payment and Real-Time Communication Systems
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1. EXECUTIVE SUMMARY

HealthLink is a sophisticated, multi-layered telemedicine ecosystem that revolutionizes healthcare delivery through the integration of Artificial Intelligence (AI), Machine Learning (ML), Real-Time Communication Technologies, Mobile Money Payment Processing, and SMS Gateway Integration. Unlike basic telemedicine applications that merely connect patients with doctors, HealthLink implements an intelligent end-to-end healthcare workflow that begins with AI-assisted symptom analysis and concludes with prescription management and follow-up care.

The platform addresses the critical healthcare access gap in Kenya and similar developing nations by implementing:

- Dual-Mode AI Triage System: Combining Machine Learning classification algorithms with Natural Language Processing (NLP) chatbot for intelligent symptom assessment
- WebRTC-Based Video Consultation: Peer-to-peer encrypted video calls with signaling server implementation
- Safaricom M-Pesa Daraja API Integration: STK Push payment initiation with callback handling for real-time payment verification
- SMS Gateway Integration: Real-time SMS notifications for offline users when messages, appointments, or prescriptions are created
- Electronic Prescription Management: Digital prescription generation with medication database and dispensing workflow
- Administrative Intelligence Dashboard: Real-time analytics, revenue tracking, and doctor verification system

Technical Complexity Highlights:

- 10+ interconnected Django application modules
- Machine Learning model using TF-IDF Vectorization and Random Forest Classification
- WebRTC signaling implementation with ICE candidate exchange
- OAuth 2.0 authentication with M-Pesa Daraja API
- Real-time notification system with SMS fallback
- Role-based access control (RBAC) for patients, doctors, and administrators

2. INTRODUCTION

2.1 Background and Context

The healthcare sector in Kenya faces significant challenges that impede effective healthcare delivery:

- Doctor-to-Patient Ratio: Kenya has approximately 1 doctor per 6,000 patients, far below the WHO recommended ratio of 1:1,000
- Geographic Disparities: Over 70% of medical specialists are concentrated in Nairobi and Mombasa, leaving rural populations underserved
- Infrastructure Limitations: Many healthcare facilities lack modern appointment management systems, resulting in 2-4 hour average wait times
- Payment Barriers: Traditional cash-based payment systems create friction in healthcare transactions

The rapid adoption of mobile technology in Kenya (95% mobile penetration) and mobile money services (over 30 million M-Pesa users) presents a unique opportunity to bridge this healthcare access gap through digital innovation.

2.2 Problem Definition

The current healthcare delivery model suffers from:

Problem	Impact	Proposed Solution
No pre-consultation symptom assessment	Patients often consult wrong specialists	AI-powered triage system
Manual appointment scheduling	Long queues and double-bookings	Digital scheduling with availability management
Cash-only payments	Payment verification delays	M-Pesa integration with instant confirmation
Paper prescriptions	Lost/illegible prescriptions, no tracking	Electronic prescription management
No communication channel	Lost follow-up, poor continuity of care	Real-time messaging + SMS notifications
Unverified practitioners	Risk of consulting unqualified individuals	Doctor verification workflow

2.3 Research Questions

- How can Artificial Intelligence be effectively utilized to perform preliminary symptom triage and recommend appropriate medical specialties?
- What architectural patterns enable seamless integration of mobile money payment systems in healthcare applications?
- How can real-time communication be maintained with offline users through SMS gateway integration?
- What security measures are necessary to protect sensitive medical data in a telemedicine platform?

2.4 Project Significance

This project contributes to:

- Sustainable Development Goal 3: Good Health and Well-being
- Kenya Vision 2030: Digital transformation of healthcare services
- Universal Health Coverage (UHC): Reducing barriers to healthcare access

3. LITERATURE REVIEW

3.1 Telemedicine Evolution and Current State

Telemedicine has evolved through distinct generations:

Generation	Era	Characteristics	Limitations
First	1960s-1990s	Video conferencing, store-and-forward	Expensive equipment, limited bandwidth
Second	2000s-2010s	Web-based platforms, EHR integration	No mobile optimization, limited interactivity
Third	2015-present	Mobile-first, AI integration, real-time	Limited adoption in developing countries
Fourth (Proposed)	Future	AI triage, integrated payments, SMS fallback	HealthLink addresses this generation

3.2 Artificial Intelligence in Healthcare

Machine Learning for Medical Triage:

Studies by Semigran et al. (2015) evaluated symptom checker accuracy, finding that correct diagnosis was achieved in only 34% of cases. However, recent advances in NLP and classification algorithms have improved accuracy significantly.

Our Approach:

HealthLink implements a hybrid triage system:

- Form-Based Triage: Structured symptom selection using predefined categories
- Chatbot Triage: Natural language processing for conversational symptom gathering
- ML Classification: TF-IDF Vectorization combined with Random Forest Classifier

```
Technical Implementation:
- Feature Extraction: TF-IDF (Term Frequency-Inverse Document Frequency)
- Classification Algorithm: Random Forest (ensemble of decision trees)
- Training Data: 100+ symptom-specialty mappings across 15 specialties
- Confidence Scoring: Probability-based specialty recommendations
```

3.3 Mobile Money Integration in Healthcare

M-Pesa has transformed financial transactions in Kenya. Integration in healthcare settings offers:

- Instant payment verification
- Transaction audit trails
- Reduced cash handling risks
- Accessibility for unbanked populations

Daraja API Technical Flow:

```
1. Patient initiates payment ->
2. System calls Daraja OAuth endpoint ->
3. Access token obtained ->
4. STK Push request sent ->
5. M-Pesa prompt appears on patient's phone ->
6. Patient enters PIN ->
```

```
7. Transaction processed ->
8. Callback URL receives confirmation ->
9. Appointment status updated automatically
```

3.4 Real-Time Communication in Healthcare

WebRTC (Web Real-Time Communication) enables peer-to-peer communication without plugins. Key components:

- Signaling Server: Exchange of session descriptions and ICE candidates
- STUN/TURN Servers: NAT traversal for establishing connections
- Media Streams: Audio/video transmission

3.5 SMS as Fallback Communication Channel

In regions with limited internet connectivity, SMS serves as a reliable fallback:

- 98% of mobile users can receive SMS
- No internet connection required
- Immediate delivery notification

Africa's Talking API and Twilio provide robust SMS gateway services suitable for healthcare notifications.

3.6 Gap Analysis

Existing System	Strengths	Weaknesses	HealthLink Advantage
Teladoc	Global presence, established	Not in Kenya, no M-Pesa	Local payment integration
Babylon Health	Strong AI	No local support	Kenyan context adaptation
MyDawa	Pharmacy focus	No consultations	Full telemedicine suite
Ponea Health	Local presence	Limited features	Comprehensive AI + payments

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4. PROPOSED SYSTEM ARCHITECTURE

4.1 System Overview

HealthLink implements a Modular Monolith Architecture using Django's application structure, enabling:

- Independent module development
- Shared database for data consistency
- Easy future migration to microservices

4.2 High-Level Architecture Diagram

```
+------------------------------------------------------------------------------+
|                              CLIENT LAYER                                     |
+------------------------------------------------------------------------------+
|  +--------------+  +--------------+  +--------------+  +--------------+    |
|  |   Patient    |  |    Doctor    |  |    Admin     |  |   Mobile     |    |
|  |   Portal     |  |   Portal     |  |  Dashboard   |  |   Browser    |    |
```

```
|  +--------------+  +--------------+  +--------------+  +--------------+    |
+--------------------------------------------------------------------------+
                                     |
                                     v
+--------------------------------------------------------------------------+
|                          APPLICATION LAYER                                |
+--------------------------------------------------------------------------+
|  +---------+ +---------+ +---------+ +---------+ +---------+ +---------+  |
|  |  Users  | |Appoint- | | Triage  | |Messaging| |Payments | |Prescrip-|  |
|  | Module  | |  ments  | |  (AI)   | | + Video | | (M-Pesa)| |  tions  |  |
|  +---------+ +---------+ +---------+ +---------+ +---------+ +---------+  |
|  +---------+ +---------+ +---------------------------------------------+ |
|  |Notifica-| | Admin   | |                                             | |
|  |  tions  | | Module  | |          Django ORM / Business Logic        | |
|  +---------+ +---------+ +---------------------------------------------+ |
+--------------------------------------------------------------------------+
                                     |
                                     v
+--------------------------------------------------------------------------+
|                          INTEGRATION LAYER                                |
+--------------------------------------------------------------------------+
|  +----------------+  +----------------+  +----------------+              |
|  |   M-Pesa       |  |     SMS        |  |    WebRTC      |              |
|  |  Daraja API    |  |   Gateway      |  |  Signaling     |              |
|  |  (Payments)    |  | (Africa's Talk)|  |   (PeerJS)     |              |
|  +----------------+  +----------------+  +----------------+              |
|  +----------------+  +----------------+                                  |
|  |   OpenAI       |  |    ML Model    |                                  |
|  |   GPT API      |  |  (Scikit-learn)|                                  |
|  |  (Chatbot)     |  |   (Triage)     |                                  |
|  +----------------+  +----------------+                                  |
+--------------------------------------------------------------------------+
                                     |
                                     v
+--------------------------------------------------------------------------+
|                            DATA LAYER                                     |
+--------------------------------------------------------------------------+
|  +--------------------------------------------------------------------+   |
|  |                  PostgreSQL / SQLite Database                       |   |
|  |  +---------+ +---------+ +---------+ +---------+ +---------+       |   |
|  |  |  Users  | |Appoint- | | Triage  | |Messages | | Payment |       |   |
|  |  | Profiles| |  ments  | |Sessions | |  Calls  | |  Trans  |       |   |
|  |  +---------+ +---------+ +---------+ +---------+ +---------+       |   |
|  |  +---------+ +---------+ +---------+                                |   |
|  |  |Prescrip-| |Notifica-| | Doctor  |                                |   |
|  |  |  tions  | |  tions  | |Availabi-|                                |   |
|  |  |         | |         | |  lity   |                                |   |
|  |  +---------+ +---------+ +---------+                                |   |
|  +--------------------------------------------------------------------+   |
+--------------------------------------------------------------------------+
```

4.3 Module Descriptions

Module	Purpose	Key Features
Users	User management	Patient/Doctor/Admin roles, Profile management, Authentication
Appointments	Scheduling	Doctor availability, Time slot booking, Status tracking

Triage

AI symptom assessment

ML model, Chatbot, Specialty recommendation

Messaging

Communication

	Real-time chat, Video calls, SMS notifications

Payments

Financial transactions

		M-Pesa STK Push, Payment verification, Transaction history

Prescriptions

Medication management

	Digital prescriptions, Medication database, Dispensing workflow

Notifications

Alerts

	In-app notifications, Email alerts, SMS fallback

Administration

System oversight

		User management, Analytics, Doctor verification

5. DETAILED SYSTEM PROCESSES

5.1 Patient Registration and Onboarding Process

```
+------------------------------------------------------------------+
|                    PATIENT REGISTRATION FLOW                     |
+------------------------------------------------------------------+
                              |
                              v
                    +-----------------+
                    |  Access Website |
                    +--------+--------+
                              |
                              v
                    +-----------------+
                    | Click "Register |
                    |   as Patient"   |
                    +--------+--------+
                              |
                              v
            +-----------------------------------+
            |      Enter Personal Details       |
            |  - Full Name                      |
            |  - Email Address                  |
            |  - Phone Number (M-Pesa)          |
            |  - Date of Birth                  |
            |  - Password                       |
            +----------------+----------------+
                              |
                              v
            +-----------------------------------+
            |    Complete Medical Profile       |
            |  - Blood Type                     |
            |  - Known Allergies                |
            |  - Medical History                |
            |  - Emergency Contact              |
            +----------------+----------------+
                              |
                              v
                    +-----------------+
                    | Account Created |
                    |   + Welcome     |
                    |   Notification  |
                    +--------+--------+
                              |
                              v
                    +-----------------+
                    |  Redirect to    |
                    |   Dashboard     |
                    +-----------------+
```

Technical Implementation:

- Django Forms with validation
- Password hashing using PBKDF2
- Automatic PatientProfile creation via Django signals
- Session-based authentication

5.2 Doctor Registration and Verification Process

```
+------------------------------------------------------------------+
|                 DOCTOR REGISTRATION & VERIFICATION                |
+------------------------------------------------------------------+
STAGE 1: REGISTRATION
---------------------
Doctor -> Submit Registration Form
          |
          +-- Personal Details (Name, Email, Phone)
          +-- Professional Details:
```

5.3 AI-Powered Symptom Triage Process

```
+------------------------------------------------------------------+
|              AI TRIAGE SYSTEM - DUAL MODE OPERATION               |
+------------------------------------------------------------------+
                         +-----------------+
                         |  Patient Starts |
                         |  Symptom Check  |
                         +--------+--------+
                                  |
                    +-------------+-------------+
                    |                           |
                    v                           v
         +-----------------+         +-----------------+
         |   FORM-BASED    |         |   AI CHATBOT    |
         |     TRIAGE      |         |     TRIAGE      |
         +--------+--------+         +--------+--------+
                  |                           |
                  v                           v
    +------------------------+   +------------------------+
    | Select Symptom Category |   | "Describe your symptoms |
```

ML Model Technical Details:

```
# Training Pipeline
1. Load symptom-specialty dataset (100+ mappings)
2. Preprocess text (lowercase, remove punctuation)
3. TF-IDF Vectorization (max_features=1000)
4. Train RandomForestClassifier (n_estimators=100)
5. Save model using joblib
# Prediction Pipeline
1. Receive user symptoms
2. Preprocess input
3. Vectorize using trained TF-IDF
4. Predict using Random Forest
5. Return top 3 specialties with confidence scores
```

5.4 Appointment Booking and Payment Process

```
+------------------------------------------------------------------+
|            APPOINTMENT BOOKING WITH M-PESA PAYMENT                |
+------------------------------------------------------------------+
STEP 1: DOCTOR SELECTION
------------------------
Patient -> Browse Doctors by Specialty
          |
          +-- Filter by: Specialization, Rating, Fee, Availability
          |
```

```
          +-- View Doctor Profile:
```

5.5 Real-Time Communication Process

```
+------------------------------------------------------------------+
|           REAL-TIME MESSAGING WITH SMS FALLBACK                   |
+------------------------------------------------------------------+
SCENARIO: Doctor sends message to Patient
    +-------------+                        +-------------+
    |   DOCTOR    |                        |   PATIENT   |
    |  (Online)   |                        |  (Offline)  |
    +------+------+                        +------+------+
           |                                      |
           |  1. Type and send message            |
           |  "Your test results are ready.       |
           |   Please schedule a follow-up."      |
           |                                      |
           v                                      |
    +-------------------------------------------------------------+
    |                    HEALTHLINK SERVER                          |
    +-------------------------------------------------------------+
    |                                                               |
    |  2. Save message to database                                  |
    |     Message {                                                 |
    |       conversation: 42,                                       |
    |       sender: "dr_amina",                                     |
    |       content: "Your test results...",                        |
    |       timestamp: "2026-01-30 10:15:00",                      |
    |       is_read: false                                          |
    |     }                                                         |
    |                                                               |
    |  3. Create in-app notification                                |
    |                                                               |
    |  4. Check patient's online status                             |
    |     +-- Patient OFFLINE                                       |
    |                                                               |
    |  5. Trigger SMS notification                                  |
    |     +-----------------------------------+                    |
    |     |  SMS GATEWAY (Africa's Talking)     |                  |
    |     |                                   |                    |
    |     |  To: +254797754321                |                    |
    |     |  From: HealthLink                 |                    |
    |     |  Message: "New message from       |                    |
    |     |  Dr. Amina on HealthLink:         |                    |
    |     |  'Your test results are ready...'   |                  |
    |     |  Login to view: healthlink.co.ke"   |                  |
    |     +-----------------------------------+                    |
    |                                                               |
    +------------------------------+--------------------------------+
                                   |
                                   |  6. SMS delivered
                                   v
                            +-------------+
                            |   PATIENT   |
                            |   (Phone)   |
                            +-------------+
                            |  [Phone] SMS     |
                            |  New message|
                            |  from       |
                            |  HealthLink |
```

```
                          +-------------+
```

SMS Notification Triggers:

Event	SMS to Patient	SMS to Doctor
Appointment Booked	[OK]	[OK]
Appointment Confirmed	[OK]	[OK]
Appointment Cancelled	[OK]	[OK]
New Message (if offline)	[OK]	[OK]
Prescription Created	[OK]	-
Payment Received	[OK]	[OK]
Video Call Incoming	[OK]	[OK]
Appointment Reminder (1hr before)	[OK]	[OK]

5.6 Video Consultation Process

```
+------------------------------------------------------------------+
|              WEBRTC VIDEO CONSULTATION FLOW                      |
+------------------------------------------------------------------+
INITIATOR: Doctor                    RECEIVER: Patient
    +-------------+                  +-------------+
    |   DOCTOR    |                  |   PATIENT   |
    |  Browser    |                  |   Browser   |
    +------+------+                  +------+------+
           |                                |
           |  1. Click "Start Video Call"   |
           |                                |
           v                                |
    +---------------------------------------------------------------+
    |                    SIGNALING SERVER                           |
    +---------------------------------------------------------------+
    |                                                               |
    |  2. Create VideoCall record                                   |
    |     {                                                         |
    |       room_id: "hl-call-abc123",                             |
    |       caller: doctor_id,                                      |
    |       receiver: patient_id,                                   |
    |       status: "initiated"                                     |
    |     }                                                         |
    |                                                               |
    |  3. Generate SDP Offer (Session Description Protocol)         |
    |     - Video codec: VP8/VP9                                    |
    |     - Audio codec: Opus                                       |
    |     - ICE candidates                                          |
    |                                                               |
    +----------------------------------------------------------------+
           |                                |
           |  4. Store caller_offer         |
           |  5. Notify patient             |
           |-------------------------------->
           |                                |
           |                         6. Patient sees
           |                         "Incoming Call from
           |                          Dr. Amina"
           |                                |
           |                         7. Click "Accept"
           |                                |
           |<--------------------------------
```

```
          |  8. Generate SDP Answer        |
          |  9. Exchange ICE candidates    |
          |                                |
    +---------------------------------------------------------------+
    |                    PEER-TO-PEER CONNECTION                    |
    +---------------------------------------------------------------+
    |                                                               |
    |  10. ICE Candidate Exchange                                   |
    |      STUN Server: stun.l.google.com:19302                    |
    |                                                               |
    |  11. Connection Established                                   |
    |      Status: "ongoing"                                        |
    |                                                               |
    |  12. Media Streams Active                                     |
    |      +---------+        +---------+                         |
    |      | Doctor  |<------>| Patient |                         |
    |      | Camera  |  P2P   | Camera  |                         |
    |      | Audio   |        | Audio   |                         |
    |      +---------+        +---------+                         |
    |                                                               |
    |  13. Call ends                                                |
    |      Duration recorded: 25 minutes                            |
    |      Status: "ended"                                          |
    |                                                               |
    +----------------------------------------------------------------+
```

5.7 Electronic Prescription Process

```
+-------------------------------------------------------------------+
|              ELECTRONIC PRESCRIPTION WORKFLOW                     |
+-------------------------------------------------------------------+
STEP 1: DOCTOR CREATES PRESCRIPTION
------------------------------------
Doctor -> Prescriptions -> Create New
          |
          +-- Select Patient (from appointments)
          |
          +-- Add Medications:
          |   +-----------------------------------------+
          |   |  Medication Database Search             |
          |   |  [Search] "Para..."                           |
          |   |                                         |
          |   |  Results:                               |
```

6. TECHNICAL IMPLEMENTATION DETAILS

6.1 Technology Stack

Layer	Technology	Justification
Backend Framework	Django 4.x (Python)	Robust ORM, built-in security, rapid development
Frontend	HTML5, CSS3, Bootstrap 5, JavaScript	Responsive design, cross-browser compatibility
Database	PostgreSQL / SQLite	Reliable, ACID compliant, scalable
Machine Learning	Scikit-learn, TF-IDF, Random Forest	Proven classification algorithms

NLP/Chatbot

OpenAI GPT-3.5 / Local LLM

Natural language understanding

Video Calls

WebRTC, PeerJS

	Peer-to-peer, low latency

Payments

M-Pesa Daraja API

	Local mobile money integration

SMS Gateway

Africa's Talking API

	Reliable Kenya SMS delivery

Web Server

```
Gunicorn + Nginx
```

	Production-grade serving

6.2 Database Schema (Key Tables)

```
-- Core User Tables
CustomUser (
    id, username, email, phone_number, user_type,
    date_of_birth, is_active, date_joined
)
DoctorProfile (
    user_id, license_number, specialization,
    years_of_experience, consultation_fee,
    is_verified, verification_status, verified_by
)
PatientProfile (
    user_id, blood_type, allergies,
    medical_history, emergency_contact
)
-- Appointment Management
DoctorAvailability (
    doctor_id, day_of_week, start_time,
    end_time, slot_duration, is_available
)
Appointment (
    id, patient_id, doctor_id, specialty_id,
    appointment_date, symptoms, status,
    created_at, updated_at
)
-- Triage System
TriageSession (
    id, user_id, symptoms, severity,
    predicted_specialty, confidence_score,
    triage_type, created_at
)
-- Communication
Conversation (
    id, patient_id, doctor_id,
    appointment_id, created_at
)
Message (
    id, conversation_id, sender_id,
    content, timestamp, is_read
)
VideoCall (
    id, room_id, caller_id, receiver_id,
    status, started_at, ended_at, duration
)
-- Payments
MpesaTransaction (
    id, user_id, appointment_id, amount,
    phone_number, mpesa_receipt_number,
    status, checkout_request_id, created_at
)
-- Prescriptions
Prescription (
    id, prescription_number, doctor_id, patient_id,
    diagnosis, notes, status, created_at
)
PrescriptionItem (
    id, prescription_id, medication_id,
    dosage, frequency, duration, instructions
)
```

```
-- Notifications
Notification (
    id, user_id, notification_type, title,
    message, link, is_read, created_at
)
SMSLog (
    id, recipient_phone, message, status,
    gateway_response, sent_at
)
```

6.3 API Endpoints

Endpoint	Method	Purpose
`/api/auth/login/`	POST	User authentication
`/api/auth/register/`	POST	User registration
`/api/triage/analyze/`	POST	ML symptom analysis
`/api/triage/chat/`	POST	Chatbot interaction
`/api/doctors/`	GET	List verified doctors
`/api/doctors/{id}/availability/`	GET	Doctor's available slots
`/api/appointments/`	GET, POST	Manage appointments
`/api/payments/mpesa/stk-push/`	POST	Initiate M-Pesa payment
`/api/payments/mpesa/callback/`	POST	M-Pesa callback handler
`/api/messages/{conversation_id}/`	GET, POST	Conversation messages
`/api/video/initiate/`	POST	Start video call
`/api/prescriptions/`	GET, POST	Manage prescriptions
`/api/notifications/`	GET	User notifications

6.4 Security Implementation

Security Measure	Implementation
Password Hashing	Django PBKDF2 with SHA256
Session Management	Secure, HttpOnly cookies
CSRF Protection	Django middleware, token validation
XSS Prevention	Template auto-escaping
SQL Injection	Django ORM parameterized queries
HTTPS	TLS 1.3 encryption
Input Validation	Django Forms, serializers
Rate Limiting	Django Ratelimit
API Authentication	Token-based (DRF)

7. SMS NOTIFICATION SYSTEM (NEW FEATURE)

7.1 Overview

The SMS notification system ensures patients and doctors receive critical updates even when they are not actively using the HealthLink platform. This addresses the reality that users may not always have the website open or may be in areas with limited internet connectivity.

7.2 SMS Gateway Integration

Provider: Africa's Talking

- Reliable delivery across Kenyan networks
- Competitive pricing
- Delivery reports
- Two-way messaging capability

7.3 Implementation Architecture

```
+------------------------------------------------------------------+
|                    SMS NOTIFICATION SERVICE                       |
+------------------------------------------------------------------+
    +-----------------+
    |  TRIGGER EVENT  |
    |  (New Message,  |
    |  Appointment,   |
    |  Prescription)  |
    +--------+--------+
             |
             v
    +-----------------+      +-----------------+
    |  Check User's   |--No-->  Send In-App    |
    |  Online Status  |      |  Notification   |
    +--------+--------+      |  Only           |
             |               +-----------------+
             | Offline
             v
    +-----------------+
    |  Format SMS     |
    |  Message        |
    |  (160 chars)    |
    +--------+--------+
             |
             v
    +-------------------------------------+
    |      AFRICA'S TALKING API           |
    |                                     |
    |  POST /version1/messaging           |
    |  {                                  |
    |    "username": "healthlink",        |
    |    "to": ["+254797754321"],         |
    |    "message": "...",                |
    |    "from": "HealthLink"             |
    |  }                                  |
    +----------------+--------------------+
                     |
                     v
    +-----------------+
    |  Log SMS in     |
    |  Database       |
    |  (Audit Trail)  |
    +-----------------+
```

7.4 SMS Templates

Event

SMS Template

Appointment Confirmed

"HealthLink: Your appointment with Dr. {name} on {date} at {time} is confirmed. Fee: KES {amount} paid. Ref: {appointment_id}"

New Message

"HealthLink: New message from {sender}. Login to view: healthlink.co.ke/messages"

Prescription Ready

"HealthLink: Dr. {name} has issued you a prescription. View at: healthlink.co.ke/prescriptions/{id}"

Appointment Reminder

"HealthLink Reminder: Your appointment with Dr. {name} is in 1 hour ({time}). Don't forget!"

Video Call

"HealthLink: Incoming video call from {caller}. Open HealthLink to join."

8. TESTING STRATEGY

8.1 Testing Levels

Level	Description	Tools
Unit Testing	Individual function testing	pytest, Django TestCase
Integration Testing	Module interaction testing	Django TestClient
System Testing	End-to-end workflows	Selenium
UAT	User acceptance testing	Real users, feedback forms
Security Testing	Vulnerability assessment	OWASP ZAP
Performance Testing	Load and stress testing	Locust

8.2 Sample Test Cases

ID	Module	Test Case	Expected Result
TC001	Users	Patient registration with valid data	Account created, profile created
TC002	Users	Login with invalid credentials	Error message displayed
TC003	Triage	ML prediction for headache symptoms	Returns Neurology specialty
TC004	Appointments	Book available slot	Appointment created as pending
TC005	Payments	M-Pesa STK Push initiation	STK prompt on user phone
TC006	Payments	Payment callback processing	Appointment status updated
TC007	Messaging	Send message when receiver offline	SMS notification triggered
TC008	Video	WebRTC connection establishment	Peer-to-peer call connected
TC009	Prescriptions	Create prescription with medications	Prescription saved, patient notified
TC010	Admin	Verify doctor credentials	Doctor status updated to verified

9. PROJECT TIMELINE

Phase	Duration	Activities
Phase 1: Research & Design	Week 1-2	Literature review, requirements gathering, system design
Phase 2: Core Development	Week 3-6	User management, authentication, basic UI
Phase 3: AI Module	Week 7-8	ML model training, chatbot integration
Phase 4: Payment Integration	Week 9-10	M-Pesa API, payment workflow
Phase 5: Communication	Week 11-12	Messaging, video calls, SMS gateway
Phase 6: Advanced Features	Week 13-14	Prescriptions, admin dashboard
Phase 7: Testing	Week 15-16	Unit, integration, UAT testing
Phase 8: Documentation	Week 17	User manuals, technical docs

Phase 9: Deployment

Week 18

		Production deployment, final presentation

10. EXPECTED OUTCOMES

10.1 Technical Deliverables

- Fully functional web application with 10+ integrated modules
- Trained ML model for symptom triage
- Integrated M-Pesa payment system
- WebRTC video consultation capability
- SMS notification gateway integration
- Comprehensive documentation

10.2 Key Performance Indicators

Metric	Target
System Uptime	> 99%
Page Load Time	< 3 seconds
ML Triage Accuracy	> 80%
Payment Success Rate	> 95%
SMS Delivery Rate	> 98%
User Satisfaction	> 4/5 rating

10.3 Impact Assessment

- Healthcare Access: Reduced hospital visits, access to specialists regardless of location
- Efficiency: Streamlined scheduling, digital prescriptions, automated payments
- Quality: Verified providers, structured feedback, consistent documentation
- Innovation: Demonstration of AI application in healthcare for Kenyan context

11. CONCLUSION

HealthLink represents a comprehensive, technically sophisticated solution to healthcare delivery challenges in Kenya. By integrating Artificial Intelligence for symptom triage, mobile money payments for accessibility, WebRTC for real-time communication, and SMS gateways for reliable notifications, the platform creates a complete digital healthcare ecosystem.

The project demonstrates practical application of:

- Machine Learning classification algorithms
- Third-party API integration (M-Pesa, SMS gateways)
- Real-time communication technologies (WebRTC)
- Modern web development frameworks (Django)
- Security best practices for healthcare data

The addition of SMS notification capabilities ensures that the platform remains accessible and functional even in scenarios where internet connectivity is limited, addressing a real-world constraint in the Kenyan context.

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